

Chapter 1

Preliminaries

1.1 Basic Covering Properties

We use the usual terms and notations^[1]. An ordinal is the set of all smaller ordinals, and a cardinal is an initial ordinal. We use κ, μ for cardinals; λ a for infinite cardinal; $\alpha, \beta, \gamma, \theta$ for ordinal numbers. $\omega = \{0, 1, 2, \dots\}$ is the smallest infinite cardinal. $|A|$ denotes the cardinality of a set A . The empty set ϕ is the unique set B such that $|B| = 0$. $\mathbb{R}$ denotes the real line $[0, 1]$ the closed unit interval in $\mathbb{R}$.

For any sets Z, Y , we denote $\mathcal{P}(Z) = \{A : A \subset Z\}$; $Y^X = \{f : f \text{ is a function from } X \text{ into } Y\}$. If $f \in Y^X$, we write $f : X \rightarrow Y$.

If $f : X \rightarrow Y$, $A \subset X$ and $B \subset Y$, then $f(A) : A \rightarrow Y$ is defined by $f|_A(x) = f(x)$ for each $x \in A$; $f[A] = \{f(x) : x \in A\}$. $f^{-1}B = f^{-1}[B] = \{x \in X : f(x) \in B\}$.

For a set A and $n < \omega$, let $[A]^n = \{B \subset A : |B| = n\}$, $[A]^{<\omega} = \cup\{[A]^n : n < \omega\}$. Let $\omega^0 = \{\phi\}$. For $n = 1, \dots, \omega$, let $\omega^n = \{f : f; n \rightarrow \omega\} = \{(k_0, k_1, \dots, k_{n-1}) : k_i \in \omega, i < n\}$. If $h < \omega$, let $\phi \oplus h = (h)$, $(k_0, \dots, k_{n-1}) \oplus h = (k_0, \dots, k_{n-1}, h)$. Let $\omega^{<\omega} = \cup_{n < \omega} \omega^n$.

Let $\mathcal{U} = \{U_\alpha \in \Delta\}$ be any family of subsets of X . Without loss of generality we may assume the indexing set Δ be any set of ordinals. We call

cardinality $|a|$ the power of $\mathcal{U}$, A λ -family is a family with power $\leq \lambda$. $\mathcal{U}$ is a cover of X if $\bigcup \mathcal{U} = X$. A λ -cover is a cover with power $\leq \lambda$.

Let $\mathcal{U} = \{U_\alpha : \alpha \in \Delta\}$, $A \subset X$, $x \in X$. We denote

$$\mathcal{U}/A = \{U_\alpha \cap A : \alpha \in \Delta\}.$$

$$(\mathcal{U})_A = \{U_\alpha : \alpha \in \Delta, U_\alpha \cap A \neq \emptyset\}; \quad (\mathcal{U})_x = \{U_\alpha : \alpha \in \Delta, x \in U_\alpha\},$$

$$st(A, \mathcal{U}) = \bigcup (\mathcal{U})_A, \quad st(x, \mathcal{U}) = \bigcup (\mathcal{U})_x,$$

$$ind(\mathcal{U})_A = \{\alpha \in \Delta : U_\alpha \cap A \neq \text{emptyset}\}; \quad quadind(\mathcal{U})_x = \{\alpha \in \Delta : x \in U_\alpha\}$$

By a space we mean a topological space with no separation axioms T_i , $i = 0, 1, 2$ at all.

Throughout the following, X denotes a space. For $A \subset X$, $\bar{A}$ denotes the closure of A and A° the interior of A . For $x \in X$, let

$$topfn(x) = \{G : G \text{ is an open subset of } X \text{ with } x \in G\}.$$